SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4307

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: **Total Marks: 30**

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

I Vaishnavee Bhavani Vijayakumar (192571071) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weig htage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconception s	Poor understandin g; major misconceptio ns
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

QUESTION 1: ONLINE SYMPOSIUM PORTAL

1. Identify appropriate HTML5 semantic elements missing in the structure

Form Grouping and Metadata: The structure lacks <fieldset> and <legend> tags to semantically group the input fields and describe the purpose of the registration section.

Input Association: <label> tags are missing. Input fields must be explicitly paired with labels using the for attribute and corresponding id attributes to define what each field represents.

Document Architecture: The snippet lacks surrounding structural semantic elements such as <main>, <section>, or <header> to establish a clear hierarchy within the webpage.

2. Modify the structure to make it standards-compliant

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Symposium Registration</title>

  <!-- Modern Inter typography -->
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700&di
splay=swap" rel="stylesheet">

  <style>
    :root {
      --primary: #4f46e5;
      --primary-hover: #4338ca;
      --primary-light: rgba(79, 70, 229, 0.1);
      --surface: #ffffff;
      --text-main: #0f172a;
      --text-muted: #64748b;
      --border-color: #e2e8f0;
      --border-focus: #6366f1;
      --radius-lg: 20px;
      --radius-md: 10px;
      --radius-sm: 8px;
      --shadow-sm: 0 1px 2px 0 rgba(0, 0, 0, 0.05);
      --shadow-card: 0 20px 45px -15px rgba(15, 23, 42, 0.1), 0 0 1px 1px rgba(15, 23, 42, 0.05);
    }
```

```
    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: 'Plus Jakarta Sans', system-ui, -apple-system, BlinkMacSystemFont, "Segoe
UI", Roboto, sans-serif;
      min-height: 100vh;
      display: flex;
      align-items: center;
      justify-content: center;
      padding: 24px 16px;
      background: radial-gradient(circle at 15% 15%, #eef2ff 0%, #f8fafc 40%),
                  radial-gradient(circle at 85% 85%, #f1f5f9 0%, #f8fafc 60%);
      color: var(--text-main);
      -webkit-font-smoothing: antialiased;
    }

    main {
      width: 100%;
      max-width: 440px;
      position: relative;
    }

    /* Subtle ambient decorative blur behind the card */
    main::before {
      content: "";
      position: absolute;
      top: -15px;
      left: 10%;
      right: 10%;
      height: 100px;
      background: linear-gradient(135deg, #a5b4fc, #c4b5fd);
      filter: blur(48px);
      z-index: -1;
      opacity: 0.6;
    }

    section {
      background: var(--surface);
      border-radius: var(--radius-lg);
      padding: 40px 36px;
```

```
  box-shadow: var(--shadow-card);
  border: 1px solid rgba(255, 255, 255, 0.8);
  backdrop-filter: blur(12px);
}

header {
  margin-bottom: 28px;
  text-align: center;
}

header h2 {
  font-size: 1.65rem;
  font-weight: 700;
  letter-spacing: -0.025em;
  color: var(--text-main);
  line-height: 1.25;
}

fieldset {
  border: none;
  display: flex;
  flex-direction: column;
  gap: 20px;
}

legend {
  width: 100%;
  font-size: 0.875rem;
  font-weight: 600;
  text-transform: uppercase;
  letter-spacing: 0.05em;
  color: var(--text-muted);
  margin-bottom: 8px;
  text-align: center;
  padding: 0;
}

fieldset > div {
  display: flex;
  flex-direction: column;
  gap: 6px;
}

label {
```

```
  font-size: 0.875rem;
  font-weight: 600;
  color: #334155;
  letter-spacing: -0.01em;
}

input {
  font-family: inherit;
  width: 100%;
  padding: 12px 16px;
  font-size: 0.95rem;
  border: 1.5px solid var(--border-color);
  border-radius: var(--radius-md);
  background-color: #f8fafc;
  color: var(--text-main);
  outline: none;
  transition: all 0.2s cubic-bezier(0.4, 0, 0.2, 1);
}

input::placeholder {
  color: #94a3b8;
}

input:hover {
  border-color: #cbd5e1;
  background-color: #ffffff;
}

input:focus {
  border-color: var(--border-focus);
  background-color: #ffffff;
  box-shadow: 0 0 0 4px var(--primary-light);
}

button[type="submit"] {
  font-family: inherit;
  width: 100%;
  margin-top: 8px;
  padding: 13px 20px;
  font-size: 0.95rem;
  font-weight: 600;
  color: #ffffff;
  background: linear-gradient(180deg, #6366f1 0%, #4f46e5 100%);
  border: none;
```

```
    border-radius: var(--radius-md);
    cursor: pointer;
    box-shadow: 0 4px 12px -2px rgba(79, 70, 229, 0.35);
    transition: all 0.2s cubic-bezier(0.4, 0, 0.2, 1);
  }

  button[type="submit"]:hover {
    background: linear-gradient(180deg, #4f46e5 0%, #4338ca 100%);
    box-shadow: 0 6px 16px -2px rgba(79, 70, 229, 0.45);
    transform: translateY(-1px);
  }

  button[type="submit"]:active {
    transform: translateY(0);
    box-shadow: 0 2px 6px -1px rgba(79, 70, 229, 0.3);
  }

  @media (max-width: 480px) {
    section {
      padding: 32px 24px;
    }

    header h2 {
      font-size: 1.45rem;
    }
  }
 </style>
</head>
<body>

 <main>
  <section>
   <header>
    <h2>Symposium Registration</h2>
   </header>

   <form action="/register" method="POST">
    <fieldset>
     <legend>Participant Details</legend>

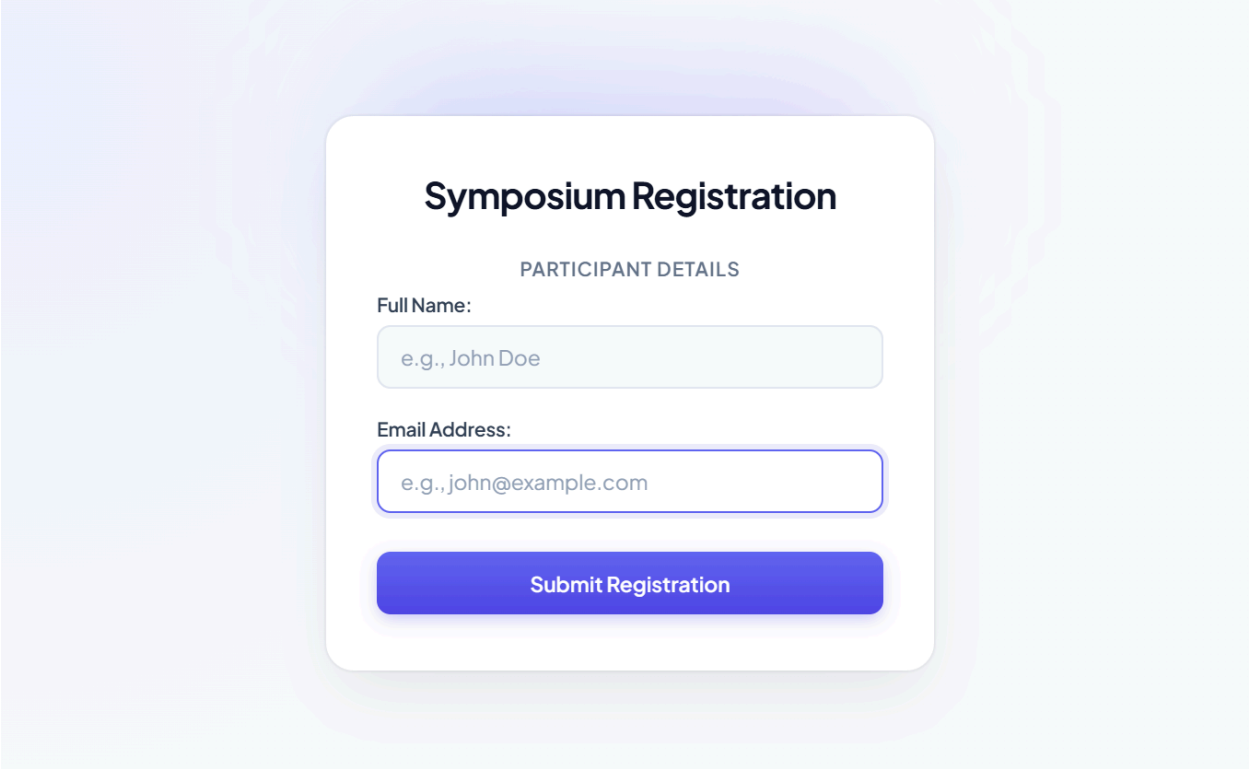
     <div>
      <label for="fullName">Full Name:</label>
      <input
       type="text"
```

```
            id="fullName"
            name="name"
            placeholder="e.g., John Doe"
            required
            autocomplete="name">
          </div>

          <div>
            <label for="userEmail">Email Address:</label>
            <input
              type="email"
              id="userEmail"
              name="email"
              placeholder="e.g., john@example.com"
              required
              autocomplete="email">
          </div>

          <div>
            <button type="submit">Submit Registration</button>
          </div>
        </fieldset>
      </form>
    </section>
  </main>

</body>
</html>
```



3. Interpret the HTTP request generated during form submission

When the user clicks the submit button, the web browser packages the form data into an HTTP POST request:

Request Line: Contains the HTTP verb (POST), the target path (/register), and the protocol version (HTTP/1.1).

Request Headers:

Host: example.com specifies the domain name of the destination server.

Content-Type: application/x-www-form-urlencoded specifies the MIME type indicating that the form keys and values are URL-encoded.

Content-Length: 35 denotes the size of the request body in bytes.

Request Body: Unlike a GET request, the payload is transmitted inside the HTTP message body separated from the headers by a blank line (CRLF). For example:

name=Jane+Doe&email=jane%40example.com

4. Analyze the HTTP response cycle from server to browser

TCP Connection & Transmission: The browser establishes a TCP handshake with the server (and TLS handshake if over HTTPS) and dispatches the raw HTTP POST request stream.

Server-Side Processing: The web server receives the request, parses the URL-encoded payload, runs validation logic, and inserts the record into the database.

Response Status Line: The server returns a status line indicating success (e.g., HTTP/1.1 200 OK or HTTP/1.1 303 See Other to redirect the browser to a confirmation page, preventing duplicate submissions on page refresh).

Response Headers: The server transmits metadata such as Content-Type: text/html; charset=UTF-8, Date, and cache control directives.

DOM Parsing and Rendering: The browser reads the HTTP response stream, constructs the Document Object Model (DOM), loads associated assets, and renders the success state to the user.

5. Suggest improvements for usability and accessibility

Explicit Labeling: Associate every <input> with a <label> using matching for and id attributes. This enables screen readers to announce the label when the field is focused and enlarges the clickable target area.

Client-Side Validation & Feedback: Add the required attribute to prevent blank submissions. Provide helpful inline error messages connected via aria-describedby and aria-invalid attributes.

Autofill Attributes: Add autocomplete="name" and autocomplete="email" to allow browsers to prefill values, speeding up registration on mobile and desktop.

Keyboard Focus Management: Ensure visible :focus outlines exist for keyboard navigation users navigating via the Tab key.

QUESTION 2: CROSS-BROWSER COMPATIBILITY ISSUE

1. Identify improper nesting and missing closing tags

Missing Closing Tags:

The heading tag <h1> is missing its closing </h1> tag.

The paragraph tag is missing its closing tag.

Improper Nesting: Because <h1> is left unclosed, the <p> element is implicitly parsed as a child of the <h1> tag inside the DOM tree, causing an invalid nesting of block-level content.

Missing HTML Boilerplate: The snippet lacks the HTML5 doctype declaration (<!DOCTYPE html>), the lang attribute on <html>, and the document <meta charset="UTF-8"> and viewport tags.

2. Analyze how different browsers may interpret this structure

Quirks Mode vs. Standards Mode: Because the <!DOCTYPE html> declaration is omitted, legacy and modern rendering engines (Blink, Gecko, WebKit) fall back into **Quirks Mode** or **Limited Quirks Mode**, emulating non-standard behavior from older browsers (e.g., box-sizing discrepancies, improper font inheritance).

Tag Soup Auto-Correction Differences:
Modern browsers use HTML5 parsing specifications to auto-close tags, but their DOM tree reconstruction can vary slightly across rendering engines. A browser might close <h1> immediately before encountering <p>, or it may nest <p> directly inside <h1>. This causes unpredictable styling rules, broken margins, and inconsistent visual layouts across browsers.

3. Identify missing semantic elements

<!DOCTYPE html>**:** Essential for enforcing W3C Standards Mode.

<meta charset="UTF-8">**:** Prevents character encoding mismatches.

<meta name="viewport" content="width=device-width, initial-scale=1.0">**:** Required for mobile responsiveness.

Structural Semantics:

<header>: To enclose introductory content and page titles.

<main>: To indicate the primary topic/content of the document.

<section> or <article>: To represent distinct thematic groupings.

4. Provide a corrected W3C-compliant HTML structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>CYBERCORP GLOBAL 2000 - Welcome</title>
```

```
<style>
 /* --- Retro 2000s Web Design System --- */
 :root {
   --win-bg: #008080; /* Classic Teal / XP Slate */
   --win-silver: #ece9d8;
   --win-card: #ffffff;
   --win-blue: #003399;
   --win-cyan: #00ffff;
   --win-header-blue: linear-gradient(180deg, #0a246a 0%, #a6caf0 100%);
   --btn-bevel-light: #ffffff;
   --btn-bevel-shadow: #7f7f7f;
   --btn-bevel-dark: #404040;
   --link-blue: #0000ee;
   --link-hover: #ff0000;
   --link-visited: #551a8b;
 }

 * {
   box-sizing: border-box;
   margin: 0;
   padding: 0;
 }

 body {
   font-family: "Tahoma", "Verdana", "Arial", sans-serif;
   font-size: 11px;
   color: #000000;
   background-color: var(--win-bg);
   background-image:
     radial-gradient(#005c5c 15%, transparent 16%),
     radial-gradient(#004c4c 15%, transparent 16%);
   background-size: 16px 16px;
   background-position: 0 0, 8px 8px;
   padding: 12px 8px 30px 8px;
   line-height: 1.4;
 }

 a {
   color: var(--link-blue);
   text-decoration: underline;
 }
 a:hover {
   color: var(--link-hover);
   text-decoration: underline;
```

```
    }
    a:visited {
      color: var(--link-visited);
    }

    /* Desktop Window Container (1024x768 optimal feel) */
    .browser-wrapper {
      max-width: 890px;
      margin: 0 auto;
      background-color: var(--win-silver);
      border: 3px outset #ffffff;
      box-shadow: 4px 6px 16px rgba(0, 0, 0, 0.45);
    }

    /* Windows XP / 2000 style Window Title Bar */
    .win-titlebar {
      background: linear-gradient(90deg, #0055ea 0%, #0866ff 40%, #6199ff 80%, #3a75e8
100%);
      color: #ffffff;
      font-weight: bold;
      font-size: 12px;
      padding: 4px 8px;
      display: flex;
      justify-content: space-between;
      align-items: center;
      text-shadow: 1px 1px 1px #002277;
      border-bottom: 2px solid #002c80;
    }

    .win-titlebar .controls {
      display: flex;
      gap: 3px;
    }

    .win-btn {
      width: 17px;
      height: 15px;
      background: #ece9d8;
      border: 1px outset #fff;
      font-size: 9px;
      font-weight: bold;
      color: #000;
      display: flex;
      align-items: center;
```

```
      justify-content: center;
      cursor: default;
      line-height: 1;
    }
    .win-btn.close {
      background: #d84937;
      color: #fff;
    }

    /* Classic Web Header Banner */
    header {
      background: linear-gradient(180deg, #001f5c 0%, #0044aa 45%, #00113a 100%);
      padding: 14px 16px 10px 16px;
      border-bottom: 2px ridge #ffffff;
      color: #ffffff;
      position: relative;
      overflow: hidden;
    }

    header::before {
      content: "";
      position: absolute;
      top: 0; left: 0; right: 0; bottom: 0;
      background: repeating-linear-gradient(0deg, rgba(255,255,255,0.04) 0px,
rgba(255,255,255,0.04) 2px, transparent 2px, transparent 4px);
      pointer-events: none;
    }

    .logo-area {
      display: flex;
      justify-content: space-between;
      align-items: flex-end;
      flex-wrap: wrap;
      gap: 8px;
    }

    .logo-title {
      font-family: "Trebuchet MS", "Impact", sans-serif;
      font-size: 28px;
      font-weight: 900;
      letter-spacing: 2px;
      color: #ffe600;
      text-shadow: 2px 2px 0px #ff0000, 4px 4px 0px #000000;
      text-transform: uppercase;
```

```
}

.logo-tagline {
  font-size: 11px;
  color: #66ffff;
  font-style: italic;
  margin-top: -2px;
  text-shadow: 1px 1px 2px #000;
}

.meta-specs {
  font-size: 10px;
  color: #d1e0ff;
  text-align: right;
  font-family: "Courier New", monospace;
}

/* Classic CSS Marquee Ticker */
.marquee-container {
  background: #000000;
  border-top: 1px solid #7f7f7f;
  border-bottom: 1px solid #7f7f7f;
  color: #00ff66;
  font-family: "Courier New", monospace;
  font-weight: bold;
  font-size: 11px;
  padding: 3px 0;
  overflow: hidden;
  white-space: nowrap;
}

.marquee-content {
  display: inline-block;
  padding-left: 100%;
  animation: marquee-roll 22s linear infinite;
}

@keyframes marquee-roll {
  0% { transform: translateX(0); }
  100% { transform: translateX(-100%); }
}

/* 2000s Beveled Navigation Bar */
nav.nav-bar {
```

```
  background: #d4d0c8;
  padding: 4px 8px;
  border-bottom: 2px groove #ffffff;
  display: flex;
  flex-wrap: wrap;
  gap: 4px;
}

.nav-tab {
  background: linear-gradient(180deg, #ffffff 0%, #ece9d8 40%, #c8c5b5 100%);
  border: 2px outset #ffffff;
  padding: 4px 10px;
  text-decoration: none;
  color: #000000;
  font-weight: bold;
  font-size: 11px;
  cursor: pointer;
  display: inline-block;
}

.nav-tab:hover {
  background: linear-gradient(180deg, #ffffe0 0%, #ffe499 50%, #f0c345 100%);
  color: #000080;
  text-decoration: none;
}

.nav-tab.active {
  background: #ffffff;
  border: 2px inset #ffffff;
  color: #003399;
}

/* 2-Column Classic Layout */
.page-body {
  display: flex;
  background: #ffffff;
  min-height: 480px;
}

@media (max-width: 680px) {
  .page-body {
    flex-direction: column;
  }
}
```

```
/* Left Sidebar */
aside.sidebar {
  width: 220px;
  background: #ece9d8;
  border-right: 2px groove #ffffff;
  padding: 10px 8px;
  flex-shrink: 0;
}

.widget-box {
  border: 2px groove #ffffff;
  background: #ffffff;
  margin-bottom: 12px;
  padding: 0;
}

.widget-title {
  background: linear-gradient(90deg, #103884 0%, #3074d6 100%);
  color: #ffffff;
  font-size: 10px;
  font-weight: bold;
  padding: 3px 6px;
  text-transform: uppercase;
  letter-spacing: 0.5px;
}

.widget-content {
  padding: 6px 8px;
  font-size: 11px;
}

.widget-content ul {
  list-style: square;
  padding-left: 16px;
  margin: 4px 0;
}

.widget-content li {
  margin-bottom: 4px;
}

/* Construction Alert Ribbon */
.under-construction {
```

```
    background: repeating-linear-gradient(-45deg, #ffd700, #ffd700 10px, #000000 10px,
#000000 20px);
    padding: 4px;
    text-align: center;
    margin-bottom: 10px;
    border: 1px solid #000;
  }

  .construction-inner {
    background: #ffffff;
    font-size: 10px;
    font-weight: bold;
    color: #cc0000;
    padding: 4px;
    text-transform: uppercase;
  }

  /* Main Content Area */
  main {
    flex: 1;
    padding: 14px 18px;
    background: #ffffff;
    background-image: linear-gradient(#f7f9fa 1px, transparent 1px);
    background-size: 100% 20px;
  }

  .welcome-banner {
    border: 2px inset #d4d0c8;
    background: #fdfcee;
    padding: 8px 12px;
    margin-bottom: 14px;
  }

  h1.main-title {
    font-size: 18px;
    color: #990000;
    font-family: "Trebuchet MS", "Arial", sans-serif;
    border-bottom: 2px solid #990000;
    padding-bottom: 2px;
    margin-bottom: 6px;
  }

  h2.section-title {
    font-size: 13px;
```

```
  color: #003399;
  background: #e6eefc;
  border-left: 5px solid #003399;
  padding: 4px 6px;
  margin: 12px 0 8px 0;
}

p.content-para {
  margin-bottom: 10px;
  line-height: 1.5;
  text-align: justify;
}

/* Feature Table */
.specs-table {
  width: 100%;
  border-collapse: collapse;
  font-size: 11px;
  margin: 10px 0;
  border: 2px outset #ffffff;
}

.specs-table th {
  background: #003366;
  color: #ffffff;
  text-align: left;
  padding: 4px 6px;
  border: 1px solid #ffffff;
}

.specs-table td {
  padding: 4px 6px;
  border: 1px solid #d4d0c8;
  background: #ffffff;
}

.specs-table tr:nth-child(even) td {
  background: #f4f6fa;
}

/* 2000s Button Action */
.action-btn {
  background: linear-gradient(180deg, #e6f2ff 0%, #99ccff 50%, #3385ff 100%);
  border: 2px outset #ffffff;
```

```
  padding: 5px 16px;
  font-weight: bold;
  font-size: 11px;
  color: #000055;
  cursor: pointer;
  display: inline-block;
  margin: 6px 0;
}

.action-btn:active {
  border-style: inset;
  background: #3385ff;
  color: #ffffff;
}

/* Retro Counter */
.visitor-counter {
  display: inline-flex;
  background: #000000;
  border: 2px inset #999;
  padding: 2px 4px;
  gap: 2px;
}

.counter-digit {
  background: #111;
  color: #00ff33;
  font-family: "Courier New", monospace;
  font-size: 13px;
  font-weight: bold;
  padding: 1px 3px;
  border: 1px solid #333;
}

/* 88x31 Badges Row */
.badge-row {
  display: flex;
  flex-wrap: wrap;
  gap: 6px;
  margin: 12px 0;
}

.retro-badge {
  width: 88px;
```

```
  height: 31px;
  border: 1px solid #000;
  box-shadow: 1px 1px 0px #fff inset;
  display: flex;
  font-size: 9px;
  font-weight: bold;
  font-family: Arial, sans-serif;
  line-height: 1;
  cursor: default;
  user-select: none;
}

.badge-left {
  width: 50%;
  display: flex;
  align-items: center;
  justify-content: center;
  text-align: center;
  color: #fff;
  padding: 1px;
}

.badge-right {
  width: 50%;
  display: flex;
  align-items: center;
  justify-content: center;
  text-align: center;
  color: #000;
  padding: 1px;
}

.badge-netscape .badge-left { background: #006666; }
.badge-netscape .badge-right { background: #ffcc00; color: #000; }

.badge-ie .badge-left { background: #003399; }
.badge-ie .badge-right { background: #ffff99; }

.badge-valid .badge-left { background: #cc0000; }
.badge-valid .badge-right { background: #ffffff; }

/* Retro Footer */
footer {
  background: #d4d0c8;
```

```
      border-top: 2px groove #ffffff;
      padding: 10px 14px;
      text-align: center;
      font-size: 10px;
      color: #444444;
    }

    .footer-links {
      margin-bottom: 6px;
    }

    .footer-links a {
      margin: 0 4px;
    }
  </style>
</head>
<body>
  <div class="browser-wrapper">

    <!-- Windows Title Bar -->
    <div class="win-titlebar">
      <span>Microsoft Internet Explorer - [CYBERCORP GLOBAL 2000]</span>
      <div class="controls">
        <div class="win-btn">_</div>
        <div class="win-btn">&#9633;</div>
        <div class="win-btn close">&#10005;</div>
      </div>
    </div>

    <!-- Header & Company Logo -->
    <header>
      <div class="logo-area">
        <div>
          <div class="logo-title">&#9733; CyberCorp 2000 &#9733;</div>
          <div class="logo-tagline">"Synergizing the Global Information Superhighway"</div>
        </div>
        <div class="meta-specs">
          [Best viewed in 1024x768 @ 16-Bit Color]<br>
          Hosted on High-Speed T1 Line // Y2K Compliant
        </div>
      </div>
    </header>

    <!-- Scrolling News Ticker -->
```

```
    <div class="marquee-container">
      <span class="marquee-content">
        *** LATEST NEWSFLASH: CyberCorp 2000 acquires dot-com multimedia networks! ***
E-commerce gateway v2.4 released *** Remember to sign our guestbook! ***
      </span>
    </div>

    <!-- Navigation Tabs -->
    <nav class="nav-bar">
      <a href="#welcome" class="nav-tab active">&#127968; Home</a>
      <a href="#about" class="nav-tab">&#128188; About Us</a>
      <a href="#services" class="nav-tab">&#128187; Services</a>
      <a href="#guestbook" class="nav-tab">&#128214; Guestbook</a>
      <a href="#webring" class="nav-tab">&#127760; Webring</a>
    </nav>

    <!-- 2-Column Content Body -->
    <div class="page-body">

      <!-- Left Sidebar -->
      <aside class="sidebar">

        <!-- Under Construction Widget -->
        <div class="under-construction">
          <div class="construction-inner">
            &#9888; Under Construction &#9888;
          </div>
        </div>

        <!-- Quick Navigation -->
        <div class="widget-box">
          <div class="widget-title">Quick Portals</div>
          <div class="widget-content">
            <ul>
              <li><a href="#about">Corporate History</a></li>
              <li><a href="#solutions">CD-ROM Media</a></li>
              <li><a href="#intranet">Intranet Dialup</a></li>
              <li><a href="#download">Download Drivers</a></li>
            </ul>
          </div>
        </div>

        <!-- Visitor Counter -->
        <div class="widget-box">
```

```
        <div class="widget-title">Visitor Statistics</div>
        <div class="widget-content" style="text-align: center;">
          <p style="margin-bottom: 4px; font-size: 10px;">Hits since Nov 1999:</p>
          <div class="visitor-counter">
            <span class="counter-digit">0</span>
            <span class="counter-digit">0</span>
            <span class="counter-digit">4</span>
            <span class="counter-digit">8</span>
            <span class="counter-digit">2</span>
            <span class="counter-digit">9</span>
            <span class="counter-digit">1</span>
          </div>
        </div>
      </div>

      <!-- Webmaster Contact -->
      <div class="widget-box">
        <div class="widget-title">Webmaster Desk</div>
        <div class="widget-content">
          <p>Questions, broken hyperlinks or midi file requests?</p>
          <p style="margin-top: 4px;">
            <a href="mailto:webmaster@cybercorp2000.net">&#9993; Send E-Mail</a>
          </p>
        </div>
      </div>
    </aside>

    <main>

      <!-- Welcome Section -->
      <section id="welcome" class="welcome-banner">
        <h1 class="main-title">Welcome to CyberCorp Global!</h1>
        <p class="content-para">
          Greetings visitor! You have arrived at the official home page of <strong>CyberCorp Global Solutions</strong>. We specialize in cross-platform corporate solutions, intranet multimedia integration, and high-bandwidth cyberspace deployment.
        </p>
      </section>

      <!-- About Us Section -->
      <section id="about">
        <h2 class="section-title">About Our Enterprise</h2>
        <p class="content-para">
```

Founded in October 1999, CyberCorp has been at the forefront of the digital revolution. We empower small-to-medium businesses with next-generation HTML architecture, client-side scripts, and interactive Flash-ready paradigms.

<h2 class="section-title">Core Competencies & Services</h2>
<table class="specs-table">
  <thead>
    <tr>
      <th>Paradigm</th>
      <th>Description</th>
      <th>Status</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Dial-Up Acceleration</td>
      <td>56k V.90 modem data compression protocols.</td>
      <td>✔ Active</td>
    </tr>
    <tr>
      <td>Multimedia CD-ROMs</td>
      <td>Interactive Macromedia Director corporate presentations.</td>
      <td>✔ Active</td>
    </tr>
    <tr>
      <td>Y2K Readiness Audits</td>
      <td>Complete mainframe date roll-over certification.</td>
      <td>Certified</td>
    </tr>
  </tbody>
</table>

  <button class="action-btn" type="button">Download Brochure (ZIP, 1.44MB)</button>

<!-- Retro Web Badges -->
<h2 class="section-title">Certifications & Compatibility</h2>

  <!-- Netscape Badge -->

    Netscape
    NOW!

```
            </div>

            <!-- Internet Explorer 6 Badge -->
            <div class="retro-badge badge-ie">
              <div class="badge-left">MSIE 6.0</div>
              <div class="badge-right">Ready</div>
            </div>

            <!-- HTML 4.01 Valid Badge -->
            <div class="retro-badge badge-valid">
              <div class="badge-left">W3C</div>
              <div class="badge-right">HTML 4.01</div>
            </div>
          </div>
        </section>

      </main>
    </div>

    <footer>
      <div class="footer-links">
        <a href="#welcome">Home</a> |
        <a href="#about">About</a> |
        <a href="#privacy">Privacy Statement</a> |
        <a href="#disclaimer">Y2K Disclaimer</a> |
        <a href="#top">Back to Top</a>
      </div>
      <p>&copy; 1999-2001 CyberCorp Global Enterprises Inc. All Rights Reserved.</p>
      <p style="margin-top: 4px; color: #666;">Optimized for Netscape Navigator 4.7 & Internet
Explorer 5.5+</p>
    </footer>

  </div>
 </body>
</html>
```



5. Suggest techniques to ensure cross-browser compatibility

HTML/CSS Validation: Run markups through the official **W3C Nu HTML Checker** to catch unclosed tags, deprecated attributes, and syntax errors.

CSS Reset / Normalize.css: Use a standard CSS reset or Normalize.css to neutralize browser-specific default style sheets (user-agent styles) across Chrome, Firefox, Safari, and Edge.

Cross-Browser Testing: Use automated multi-browser platforms (e.g., BrowserStack, Playwright, Selenium) to test and verify rendering across multiple engines (Gecko, Blink, WebKit).

Feature Detection: Use tools like Modernizr or native CSS @supports queries instead of browser/user-agent sniffing to gracefully handle unsupported features.

QUESTION 3: FORM SUBMISSION FAILURE

1. Identify issues in the form configuration

Incorrect Button Type: The button has type="button". The default behavior of type="button" is to do nothing when clicked unless attached to a JavaScript event listener. To submit a form natively, it must be type="submit".

Insecure HTTP Method: The form uses method="GET" for sensitive authentication credentials (password), which exposes passwords in the URL.

Missing Accessibility & Form Validation Attributes: Input fields lack associated <label> tags, and fields are not marked as required.

2. Analyze why the HTTP request is not triggered

Under the W3C HTML specifications, a form submission event is automatically triggered only by an <input type="submit">, <button type="submit">, or an implicit submit button (a <button> tag without an explicit type attribute).

Because the developer specified type="button", the browser treats the element as a generic, unassigned button.

In the absence of an attached JavaScript event listener (e.g., addEventListener('click', ...) calling form.submit()), clicking the button produces no action, preventing the form submit event and network request from firing.

3. Interpret the role of GET method in this scenario

Data Appending in URL Query String: The GET method instructs the browser to append all serialized form field data to the target action URL as query parameters:

/submit?username=user1&password=mysecretpassword

Severe Security Vulnerabilities:

> **Browser History & Logs:** The password is saved in plaintext in the browser history, server access logs, and proxy logs.
>
> **Referrer Header Leakage:** The password in the URL is exposed to third-party domains via the Referer header if the user navigates to an external link.
>
> **Screen Peeking:** Plaintext passwords are displayed directly in the browser's address bar.

HTTP Idempotence Violation: In the HTTP protocol specification, GET is designated as a safe, idempotent method used strictly to retrieve data without modifying server state. Authentication modifies session state and should never use GET.

4. Propose a corrected implementation

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Account Login</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
```

```
  <link
href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700&di
splay=swap" rel="stylesheet">
  <style>
    :root {
      --bg-page: #f4f6f9;
      --card-bg: #ffffff;
      --card-border: #e2e8f0;
      --text-main: #1e293b;
      --text-muted: #64748b;
      --text-label: #334155;
      --primary: #3b82f6;
      --primary-hover: #2563eb;
      --primary-active: #1d4ed8;
      --input-border: #cbd5e1;
      --input-hover: #94a3b8;
      --input-focus-ring: rgba(59, 130, 246, 0.2);
      --shadow-sm: 0 1px 2px 0 rgba(0, 0, 0, 0.05);
      --shadow-card: 0 10px 25px -5px rgba(15, 23, 42, 0.08), 0 8px 10px -6px rgba(15, 23, 42,
0.04);
      --radius: 10px;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: 'Plus Jakarta Sans', -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto,
sans-serif;
      background-color: var(--bg-page);
      background-image:
        radial-gradient(at 0% 0%, rgba(219, 234, 254, 0.6) 0px, transparent 50%),
        radial-gradient(at 100% 100%, rgba(224, 231, 255, 0.5) 0px, transparent 50%);
      color: var(--text-main);
      min-height: 100vh;
      display: flex;
      flex-direction: column;
      align-items: center;
      justify-content: center;
      padding: 24px 16px;
      -webkit-font-smoothing: antialiased;
```

```
}

.login-container {
  width: 100%;
  max-width: 420px;
}

.brand-header {
  text-align: center;
  margin-bottom: 24px;
}

.brand-icon {
  width: 44px;
  height: 44px;
  margin: 0 auto 12px;
  background: linear-gradient(135deg, #eff6ff 0%, #dbeafe 100%);
  border: 1px solid #bfdbfe;
  border-radius: 12px;
  display: flex;
  align-items: center;
  justify-content: center;
  color: var(--primary);
  box-shadow: var(--shadow-sm);
}

.brand-title {
  font-size: 20px;
  font-weight: 700;
  color: var(--text-main);
  letter-spacing: -0.02em;
}

.brand-subtitle {
  font-size: 13px;
  color: var(--text-muted);
  margin-top: 4px;
}

form {
  background: var(--card-bg);
  border-radius: var(--radius);
  box-shadow: var(--shadow-card);
  border: 1px solid var(--card-border);
```

```
  padding: 28px 24px;
}

fieldset {
  border: 1px solid #e2e8f0;
  border-radius: 8px;
  padding: 20px 18px 18px;
  background-color: #fafbfc;
}

legend {
  font-size: 13px;
  font-weight: 600;
  letter-spacing: 0.02em;
  color: var(--primary-hover);
  padding: 0 10px;
  background: #ffffff;
  border: 1px solid #e2e8f0;
  border-radius: 20px;
  box-shadow: var(--shadow-sm);
}

fieldset > div {
  margin-bottom: 16px;
}

label {
  display: block;
  font-size: 13px;
  font-weight: 600;
  color: var(--text-label);
  margin-bottom: 6px;
}

input[type="text"],
input[type="password"] {
  width: 100%;
  height: 42px;
  font-family: inherit;
  font-size: 14px;
  color: var(--text-main);
  background-color: #ffffff;
  border: 1px solid var(--input-border);
  border-radius: 6px;
```

```
  padding: 0 12px;
  transition: border-color 0.18s ease, box-shadow 0.18s ease, background-color 0.18s ease;
  outline: none;
}

input[type="text"]::placeholder,
input[type="password"]::placeholder {
  color: #94a3b8;
}

input[type="text"]:hover,
input[type="password"]:hover {
  border-color: var(--input-hover);
}

input[type="text"]:focus,
input[type="password"]:focus {
  border-color: var(--primary);
  box-shadow: 0 0 0 3px var(--input-focus-ring);
  background-color: #ffffff;
}

fieldset > div:last-child {
  margin-bottom: 0;
  margin-top: 20px;
}

button[type="submit"] {
  width: 100%;
  height: 42px;
  font-family: inherit;
  font-size: 14px;
  font-weight: 600;
  letter-spacing: 0.01em;
  color: #ffffff;
  background-color: var(--primary);
  border: 1px solid rgba(0, 0, 0, 0.06);
  border-radius: 6px;
  cursor: pointer;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  box-shadow: 0 1px 2px rgba(37, 99, 235, 0.2);
  transition: background-color 0.16s ease, transform 0.08s ease, box-shadow 0.16s ease;
```

```
    }

    button[type="submit"]:hover {
      background-color: var(--primary-hover);
      box-shadow: 0 4px 8px rgba(37, 99, 235, 0.24);
    }

    button[type="submit"]:active {
      background-color: var(--primary-active);
      transform: translateY(1px);
      box-shadow: 0 1px 2px rgba(37, 99, 235, 0.2);
    }

    .form-footer-notes {
      margin-top: 18px;
      text-align: center;
      font-size: 12px;
      color: var(--text-muted);
    }

    .form-footer-notes a {
      color: var(--primary);
      text-decoration: none;
      font-weight: 500;
    }

    .form-footer-notes a:hover {
      text-decoration: underline;
    }
  </style>
</head>
<body>

  <div class="login-container">
    <div class="brand-header">
      <div class="brand-icon">
        <svg width="22" height="22" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2" stroke-linecap="round" stroke-linejoin="round">
          <path d="M12 22s8-4 8-10V5l-8-3-8 3v7c0 6 8 10 8 10z"/>
        </svg>
      </div>
      <h1 class="brand-title">Client Portal</h1>
      <p class="brand-subtitle">Please enter your credentials to access your dashboard</p>
    </div>
```

```
    <form action="/login" method="POST">
      <fieldset>
        <legend>Account Login</legend>

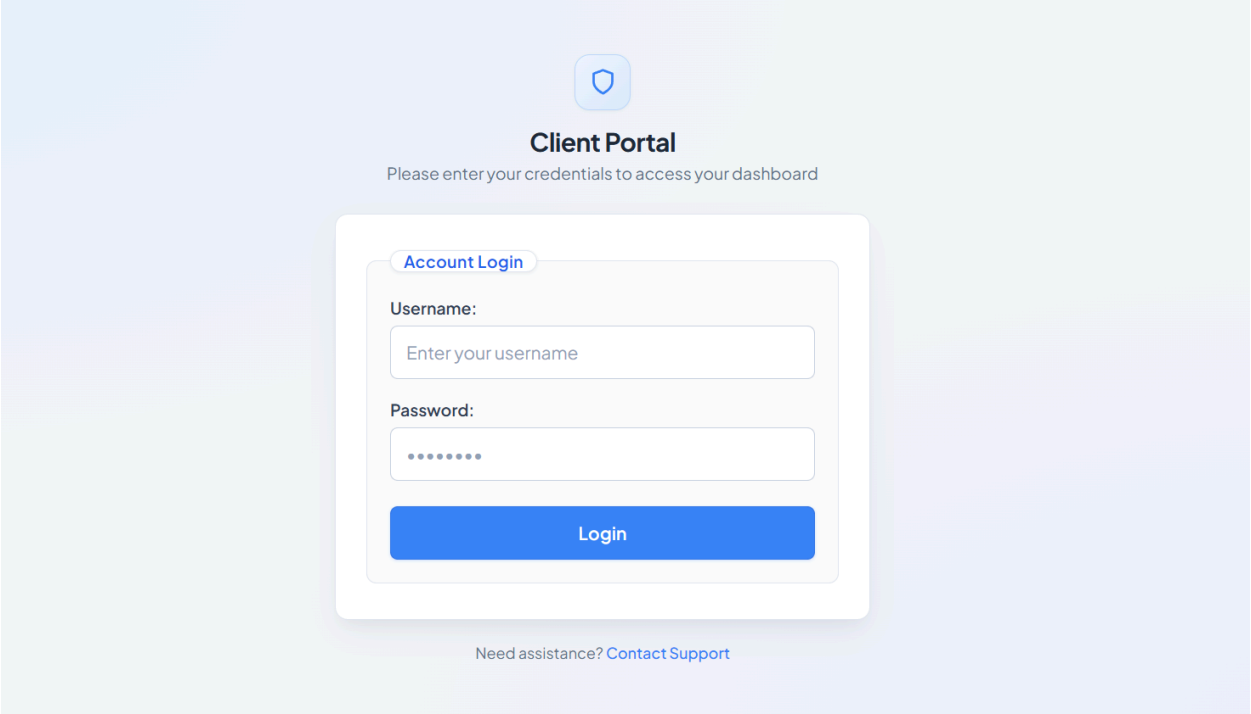
        <div>
          <label for="loginUsername">Username:</label>
          <input
            type="text"
            id="loginUsername"
            name="username"
            placeholder="Enter your username"
            required
            autocomplete="username">
        </div>

        <div>
          <label for="loginPassword">Password:</label>
          <input
            type="password"
            id="loginPassword"
            name="password"
            placeholder="••••••••"
            required
            autocomplete="current-password">
        </div>

        <div>
          <!-- Changed type from "button" to "submit" -->
          <button type="submit">Login</button>
        </div>
      </fieldset>
    </form>

    <div class="form-footer-notes">
      Need assistance? <a href="#help">Contact Support</a>
    </div>
  </div>

</body>
</html>
```



5. Suggest improvements for secure data transmission

Enforce HTTPS (TLS/SSL): Serve all pages over HTTPS to encrypt the transport layer. This prevents eavesdropping and Man-in-the-Middle (MITM) attacks during data transmission.

Switch to POST Method: Transmit authentication credentials inside the encrypted HTTP request body rather than the URL query string.

CSRF Protection: Include a cryptographically secure, randomized Anti-Cross-Site Request Forgery (CSRF) token as a hidden input field to protect against unauthorized cross-site actions.

Security Headers: Configure the web server with security headers such as Strict-Transport-Security (HSTS), X-Content-Type-Options: nosniff, and a strict Content-Security-Policy (CSP).

Rate Limiting & Password Masking: Implement server-side rate limiting on the /login endpoint to mitigate brute-force attacks and ensure passwords are never stored or logged in plain text.